Cyber security - is the practice of defending computers, servers, mobile devices, electronic systems, networks, and data from malicious attacks.

Vulnerability, Threat, Risk - A vulnerability is a flaw or weakness in an asset's design, implementation, or operation and management that could be exploited by a threat.

- A threat is a potential for a threat agent to exploit a vulnerability.
- A RISK is the potential for loss when the threat happens.

  RISK = Threat × Vulnerability.

Cryptology - is the all-inclusive term for the study of communication over nonsecure channels, and related problems.

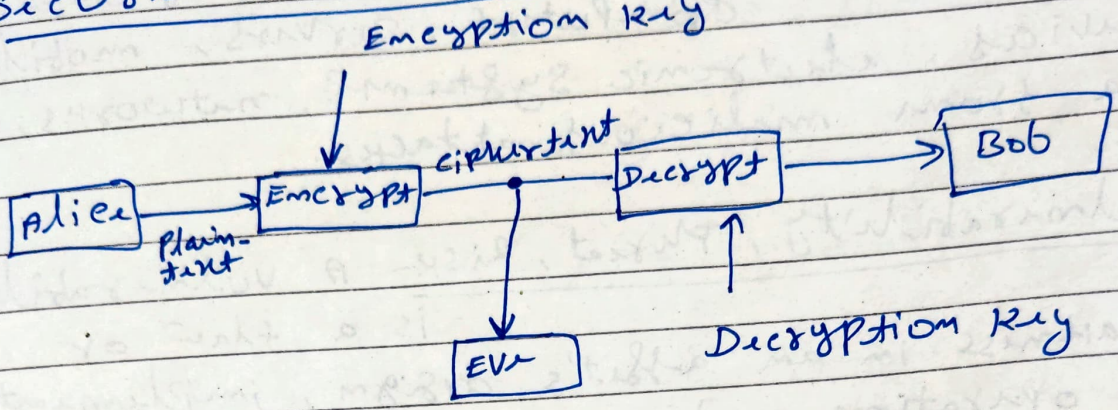
Cryptography - The process of designing systems to do this is called cryptography.

Cryptanalysis - It deals with breaking cryptographic systems.

Book - Introduction to cryptography with coding Theory.
Author - Wade Trappe & Lawrence C. W
Pub - Pearson

Date : __/__/____

Secure Communications.



- Encryption method is assumed to be known to Eve.

Eve could have one of the following goals -

(i) Read message
(ii) Find the key and thus read all messages encrypted with that key.
(iii) Alter Alice's messages
(iv) masquerade as Alice, and communicate with Bob ~~when~~ (Bob believes he is communicating with Alice).

Possible attacks - How Eve can determine the key is dependent on how much information is available with him.
Four attacks are as follows -

(i) Ciphertext only : Eve has copy of ciphertext only.
(ii) Known plaintext : Eve has plaintext and corresponding ciphertext.

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Let Alice always start message with Dear Bob,

(iii) Chosen Plaintext : Let Eve get temporary access to encryption machine but not key.

(iv) Chosen ciphertext : Eve get temporary access to decryption machine.